

Ying Fu

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Education

University of Wisconsin-Madison

Ph.D. in Industrial and Systems Engineering

Thesis: *Degradation-Aware Learning and Optimization: Modeling, Remote Prognostics, and Proactive Maintenance in Smart and Connected Manufacturing Systems*

Advisor: [Prof. Kaibo Liu](#)

Madison, WI, USA

09/2022-05/2027 (expected)

University of Wisconsin-Madison

M.S. in Computer Science

Madison, WI, USA

09/2022-12/2025

South China University of Technology

M.S. in Management Science and Engineering

Guangzhou, China

09/2019-06/2022

South China University of Technology

B.Eng. in Polymer Materials and Engineering

B.S. in Finance (Double degree)

Guangzhou, China

09/2015-06/2019

Research Interests

Methodology: degradation modeling and prognostics; machine learning and AI (deep learning, LLMs, and generative models); digital twins; reinforcement learning; optimization and data-driven decision-making.

Applications: smart manufacturing; logistics robotics; fusion facilities and nuclear energy systems.

Publications

Published

- [1] **Fu Y.**, Huh YK., and Liu K. (2026), “Dynamic Sensor Selection for Remote Prognostics.” *IJSE Transactions*, 58(9):1111-1125.
 - **Selected as a feature article in the September 2026 issue.**
- [2] **Fu Y.**, Huh YK., and Liu K. (2025), “Degradation Modeling and Prognostic Analysis Under Unknown Failure Modes.” *IEEE Transactions on Automation Science and Engineering*, 22:11012-11025.
- [3] Huh YK., **Fu Y.**, and Liu K. (2025), “A Bayesian Spike-and-Slab Sensor Selection Approach for High-Dimensional Prognostics.” *IEEE Transactions on Automation Science and Engineering*, 22:13814-13827.
- [4] Zhu W., **Fu Y.**, and Zhou Y. (2024), “3D Dynamic Heterogeneous Robotic Palletization Problem.” *European Journal of Operational Research*, 316(2):584-596.
- [5] **Fu Y.**, Liu K., and Zhu W. (2024), “Instance Selection via Voronoi Neighbors for Binary Classification Tasks.” *IEEE Transactions on Knowledge and Data Engineering*, 36(8):3921-3933.

- **Best Paper Finalist, DAIS Section, IISE Annual Conference & Expo, 2023.**

[6] Zheng R., Peng Z., **Fu Y.**, Deng Z., Liu S., Xing S., Wu Y., Li J., and Liu L. (2020), “A Novel Conductive Core-Shell Particle Based on Liquid Metal for Fabricating Real-Time Self-Repairing Flexible Circuits.” *Advanced Functional Materials*, 30(15):1910524.

Under Revision/Review

- [7] **Fu Y.**, Huh YK., and Liu K., “A Degradation-Aware Cross-Attention Framework for Multimodal Prognostics with Discrete Event Logs and Continuous Sensor Signals.” *IISE Transactions*, under revision.
- [8] Zhu W., Qiu R., and **Fu Y.**, “Comparative Study on the Performance of Categorical Variable Encoders in Classification and Regression Tasks.” *Data Mining and Knowledge Discovery*, under review.

In Preparation

- [9] **Fu Y.** and Liu K., “Fleet Maintenance and Workload Allocation under Unknown Degradation Dynamics: A Learning and Optimization Approach.” To be submitted to *Manufacturing & Service Operations Management*.
- [10] **Fu Y.** and Liu K., “Large Language Models for Extracting Operational Knowledge from Unstructured Records in Nuclear Energy Systems.” In preparation.
- [11] **Fu Y.**, Lee J., and Liu K., “Event-Based Control for Proactive Degradation Management.” In preparation.

Honors and Awards

• Grainger Wisconsin Distinguished Graduate Fellowship	2026-2027
• Feature Article Selection, <i>IISE Transactions</i>	2026
• Campus-Wide Teaching Assistant Award Nominee, Dept. of ISyE, UW-Madison	2025-2026
• IISE Future Faculty Fellow (3F), IISE	2025-2026
• Narayan-Chen Family Graduate Student Support Fund Scholarship, UW-Madison	2025
• Best Paper Finalist, DAIS Section, IISE Annual Conference & Expo	2023
• ISyE Graduate Student Travel Award, UW-Madison	2023
• Chancellor’s Opportunity Award, UW-Madison	2022
• National Scholarship, the Ministry of Education in China	2017

Contributed Grant Proposals

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- **Optimizing Performance and Security for Advanced Microreactor Fleets: Decentralized Monitoring, Collaborative Failure Prediction, and Fleet-Level Operation Control and Maintenance**
 - U.S. Department of Energy (DOE); Status: Not funded.
 - Contributed to proposal development and submission.
 - **Capability-Aware Autonomy: Coupled Planning, Control, Maintenance, and Logistics for Long-Duration Multi-Vehicle Operations**
 - Office of Naval Research (ONR); Status: Pending.
 - Contributed to proposal development and submission.

Teaching Experience

- **Course Co-Designer**, Industry Workshop, UW–Madison Summer 2026
Turning Data into Insight: A Hands-On Tableau Short Course
 - Co-developed materials for a four-day, half-day professional workshop on Tableau-based data analytics for industry professionals, including lecture slides, hands-on exercises, and projects.
- **Teaching Assistant**, Dept. of ISyE, UW–Madison Fall 2025
ISyE 412: Fundamentals of Industrial Data Analytics (evaluation: 4.67/5.00)
- **Teaching Assistant**, Dept. of ISyE, UW–Madison Spring 2025
ISyE 412: Fundamentals of Industrial Data Analytics (evaluation: 4.30/5.00)
 - Led weekly R labs, held office hours, graded coursework, and assisted with exam design.

Teaching Interests

Data Analytics and AI: industrial data analytics; machine learning; deep learning; generative AI and large language models; AI agents.

Operations Research and Optimization: linear and nonlinear optimization; data-driven optimization; dynamic programming and reinforcement learning.

Reliability, Quality, and Prognostics: reliability and quality engineering; degradation modeling; system monitoring and prognostics; predictive maintenance.

Stochastic Systems and Simulation: probability; stochastic processes; simulation modeling.

Students Mentored

- **Douglas Weittenhiller III**, Undergraduate, Dept. of ISyE, UW–Madison Spring 2025; Fall 2025
Guided the student in developing Raspberry Pi communication for an IoT testbed and supporting data collection.
Next position: Industrial Engineer at Prolec GE Waukesha
- **Joanne Yun**, Undergraduate, Dept. of ISyE, UW–Madison Spring 2026
Guided the student in reviewing and implementing federated learning methods for distributed prognostic modeling.
Next position: Graduate Student in ISyE at UW–Madison

AI/ML Workshops for Industry Professionals

- **Tutorials on Designing and Deploying AI Agents Locally**
 - Attendees: 35
 - Delivered lecture: *Querying and Analyzing Spreadsheets Through Conversations* [[Colab Material](#)] 08/2026
- **Tutorials on Supervised Learning for Industrial IoT Data**
 - Attendees: 23
 - Delivered lecture: *Tree-Based Methods: Decision Trees, Bagging, and Random Forests* [[Colab Material](#)] 05/2026
- **Tutorials on Time Series Analytics and Adaptations of Large Language Models (LLMs)**
 - Attendees: 23
 - Delivered lecture: *Adaptation of Large Language Models (LLMs)* [[Colab Material](#)] 08/2025
- **Tutorials on Methods for Complex Multivariate Time Series Data Analytics**
 - Attendees: 23
 - Delivered lecture: *Model Evaluation and Uncertainty Quantification* [[Colab Material](#)] 05/2025
- **Tutorials on Generative AI**
 - Attendees: 23
 - Delivered lecture: *Generative Adversarial Networks (GANs)* [[Colab Material](#)] 01/2025

Professional Activities

- **Reviewer**
 - IEEE Transactions on Automation Science and Engineering
 - IEEE Transactions on Knowledge and Data Engineering
 - IISE Annual Conference & Expo
- **Invited Session Chair**, INFORMS Annual Meeting 11/2026
 - *Learning to Decide: Reinforcement Learning and Large Language Models for Smart Systems Decision-Making*, San Francisco, CA
- **Invited Session Chair**, INFORMS Annual Meeting 10/2025
 - *Reinforcement Learning for Process Control and Maintenance*, Atlanta, GA
- **Vice President**, SME Student Chapter, UW-Madison 09/2022-08/2024
- **Panelist**, *Graduate Seminar in Industrial Engineering*, Dept. of ISyE, UW-Madison Fall 2025
- **Panelist**, *ISyE Student Research Symposium*, Dept. of ISyE, UW-Madison Spring 2026

Invited Presentations

- **Maintenance and Workload Allocation under Unknown Degradation Dynamics: A Learning and Optimization Approach**
 - INFORMS Annual Conference, San Francisco, CA, USA 11/2026
- **Degradation Control and Maintenance via Reinforcement Learning**
 - INFORMS Annual Conference, Atlanta, GA, USA 10/2025
- **Dynamic Sensor Selection for Remote Prognostics**
 - INFORMS Annual Conference, Seattle, WA, USA 10/2024
- **Degradation modeling and prognostic analysis under unknown failure modes**
 - INFORMS Annual Conference, Phoenix, AZ, USA 10/2023
- **Instance selection via Voronoi neighbors for binary classification tasks**
 - Industrial and Systems Engineering Research Conference (IISE), New Orleans, LA, USA 05/2023

Professional Society Memberships

- Institute for Operations Research and the Management Sciences (INFORMS)
- Institute of Industrial and Systems Engineers (IISE)
- Institute of Electrical and Electronics Engineers (IEEE)
- Society of Manufacturing Engineers (SME)

REFERENCES

Dr. Kaibo Liu

Grainger STAR Professor
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University of Wisconsin-Madison
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Assistant Professor

Department of Industrial and Systems Engineering

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Dr. Ben Lindley

Associate Professor

Department of Nuclear Engineering & Engineering Physics

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